Land Use and Land Cover Classification from Satellite Imagery

Deep Learning Course Project Proposal

Team: Abdelrahman Osman · Mohammed Alroomi

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1. Problem Definition

Satellite constellations such as the European Space Agency's Sentinel-2 now image the entire Earth's land surface every few days, producing far more optical imagery than any team of analysts could label by hand. Automating the classification of what a given patch of land is being used for — cropland, forest, urban development, water, and so on — makes it possible to track urban expansion, monitor agricultural land, detect deforestation, and support rapid land-cover mapping after natural disasters, at a scale and update frequency manual photo-interpretation cannot match.

Input: a 64×64 pixel RGB image patch (3-band optical imagery derived from Sentinel-2). **Output:** a single categorical label, one of 10 land-use / land-cover classes. **Task type:** supervised multi-class (single-label) image classification.

2. Dataset

We use **EuroSAT** (Helber, Bischke, Dengel & Borth, 2019, IEEE JSTARS; originally presented at IGARSS 2018), a benchmark dataset built from Sentinel-2 satellite imagery covering 34 European countries. It contains **27,000 labeled, geo-referenced images** across **10 classes**. We downloaded the official RGB release and inspected every file ourselves: all 27,000 images are 64×64 pixels, RGB, JPEG, matching the dataset documentation exactly. The table below shows the real per-class counts from that inspection — classes are not perfectly balanced, ranging from 2,000 to 3,000 images.

Class	# Images	Class	# Images
AnnualCrop	3,000	PermanentCrop	2,500
Forest	3,000	Residential	3,000
HerbaceousVegetation	3,000	River	2,500
Highway	2,500	SeaLake	3,000
Industrial	2,500	Pasture	2,000

Total: 27,000 images. Source: our own file-by-file inspection (scripts/inspect_data.py).

Licensing / access: the EuroSAT labels and repository are released under the MIT license; the underlying Sentinel-2 imagery is free, full and open under the EU Copernicus data policy. The dataset is freely available for academic coursework use, with attribution to Helber et al. (2019). Official repository: github.com/phelber/eurosat.

Representative samples and class distribution



Figure 1. One real, randomly sampled image per class (all classes are visually distinct at 64×64 resolution).

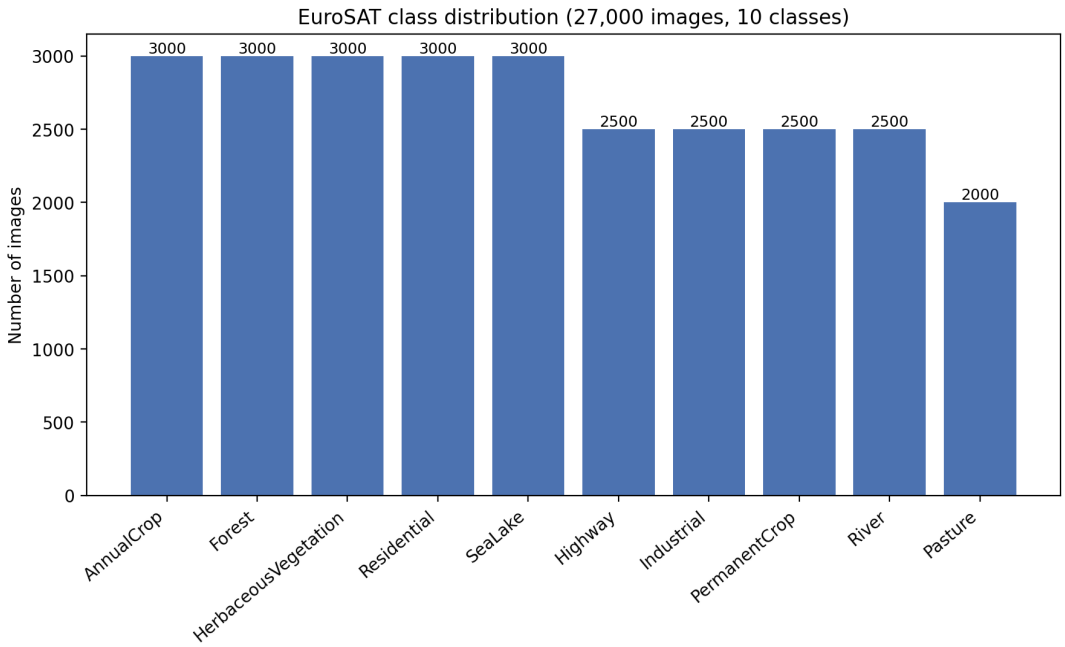


Figure 2. Class frequency distribution across the full 27,000-image dataset (real counts, not estimated).

3. Train / Validation / Test Split

EuroSAT does not ship an official split, so we define one ourselves, before any model training, stratified by class so every split preserves the same class proportions as the full dataset. A fixed random seed (42) makes the split fully reproducible (scripts/split_dataset.py).

Split	Fraction	# Images	Purpose
Train	70%	18,900	Fit model weights
Validation	15%	4,050	Model selection, hyperparameter tuning, early stopping
Test	15%	4,050	Final evaluation only, at the end of the semester

4. Validation Metric

We use **macro-averaged F1-score** on the validation set as the single metric to track and compare models throughout the semester. Because class sizes range from 2,000 to 3,000 images (a real, if moderate, imbalance), macro-F1 weighs every class equally and will not mask poor performance on smaller classes (e.g. Pasture) the way raw accuracy could. Overall accuracy will still be reported for reference, but macro-F1 is the metric used for model selection and early stopping.

5. Planned Progression Across the Semester

(1) Baseline: a small convolutional neural network trained from scratch. **(2)** Deeper CNN with regularization and data augmentation (rotations and flips are valid for overhead satellite imagery, which has no canonical "up"). **(3)** Transfer learning: fine-tune an ImageNet-pretrained backbone (e.g. ResNet-18/50) — this is close to how the original EuroSAT paper reached 98.57% accuracy. **(4)** Stretch goal: repeat the comparison using the 13-band multispectral version of EuroSAT to test whether additional Sentinel-2 spectral bands (e.g. near-infrared) improve discrimination between visually similar classes such as AnnualCrop and PermanentCrop.

References

Helber, P., Bischke, B., Dengel, A., & Borth, D. (2019). EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing.* arXiv:1709.00029. Official repository: github.com/phelber/eurosat.